CENG 467 — Natural Language Understanding and Generation

Take-Home Midterm Examination Report

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Abstract

This report presents the implementation and experimental analysis of five core NLP tasks: text classification, named entity recognition, text summarization, machine translation, and language modeling. For each task, we compare multiple modeling approaches ranging from classical machine learning to pre-trained transformer-based models. All experiments are fully reproducible with fixed random seeds (seed = 42), consistent dataset splits, and documented hyperparameters. The code and notebooks are available at the repository link above. Key findings are: [fill in after completing experiments — 2–3 sentences summarizing main results across all 5 questions].

Contents

1 Question 1: Representation Learning in Text Classification

1.1 Problem Formulation

We study binary sentiment classification on movie reviews. Given a review x, the task is to predict a label $y \in \{0, 1\}$ where 0 denotes negative sentiment and 1 denotes positive sentiment. The goal is to compare sparse, dense, and contextual representations under a consistent evaluation protocol.

1.2 Dataset Description

We use the IMDb dataset with the official train and test splits, and create a stratified 10% validation split from the training set (seed 42). The data are balanced across classes.

Table 1: IMDb dataset statistics.

Split	**Examples**	**Avg Len**	**Positive**	**Negative**
Train	22,500	233.86	11,250	11,250
Validation	2,500	233.17	1,250	1,250
Test	25,000	228.53	12,500	12,500

1.3 Methodology

Preprocessing and tokenization. We compare two tokenization strategies: (i) a classical pipeline using NLTK word tokenization with optional lowercasing and stopword removal, and (ii) the DistilBERT tokenizer for contextual modeling. For the classical pipeline, we run an ablation over four configurations (A1–A4) and pick the best validation accuracy. A truncation study shows that larger token budgets improve validation accuracy (100 tokens: 0.8444, 200 tokens: 0.8720, 400 tokens: 0.8948).

Models.

- **TF-IDF + Logistic Regression** (sparse): unigram and bigram TF-IDF features with linear classifier.
- **BiLSTM + GloVe** (dense static): word embeddings initialized with GloVe 100d, BiLSTM encoder, mean pooling, and linear output.
- **DistilBERT** (contextual): fine-tuned DistilBERT for sequence classification.

1.4 Experimental Setup

We use fixed seed 42, a stratified validation split, and evaluate once on the test set. Metrics are Accuracy and Macro-F1. Training time is recorded for each model. Confusion matrices and training curves are saved for analysis.

1.5 Results

Table 2: Test set results for Q1.

Model	**Representation**	**Accuracy**	**Macro-F1**
TF-IDF + LR	Sparse	0.8966	0.8966
BiLSTM + GloVe	Dense (static)	0.8898	0.8898
DistilBERT	Contextual	0.9296	0.9296

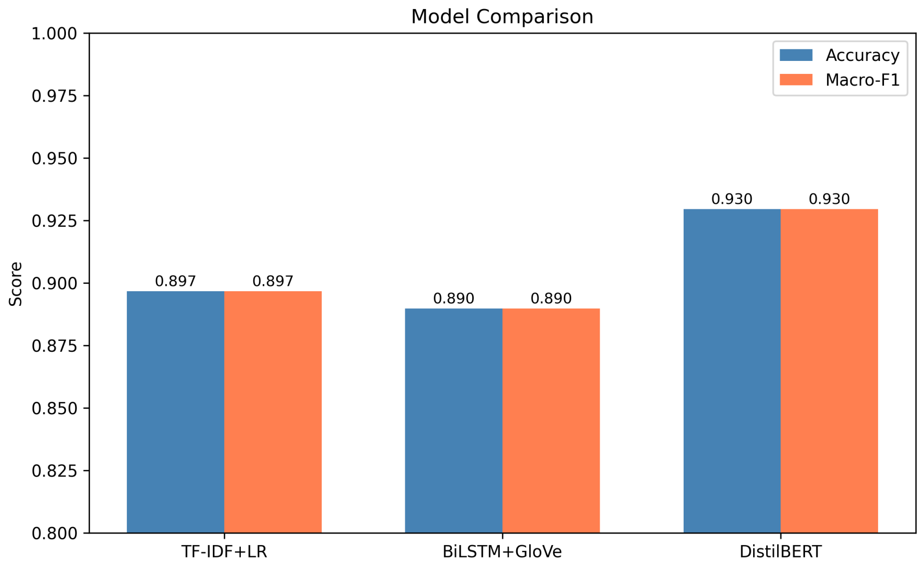
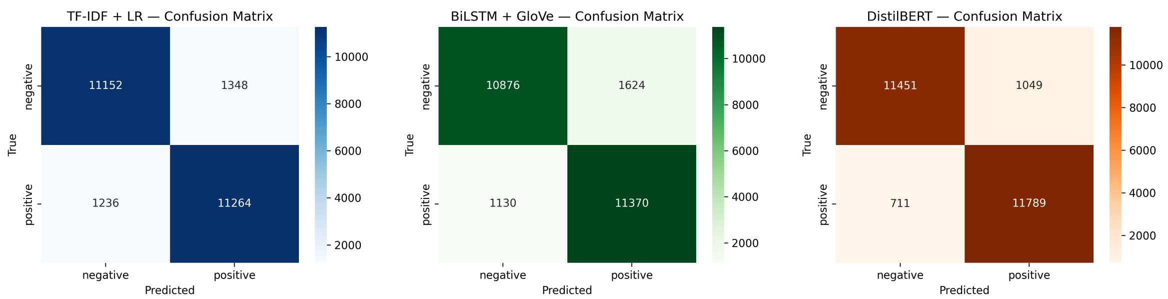
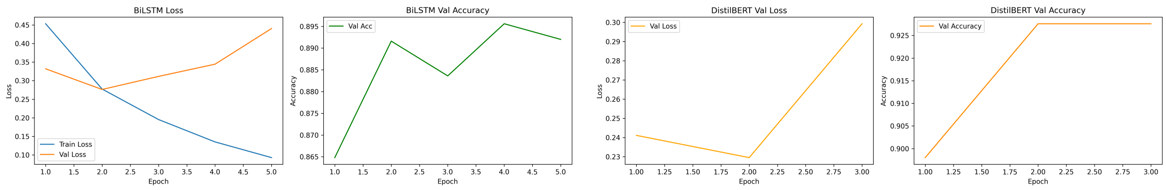


Figure 1: Accuracy and Macro-F1 comparison across models.



(a) TF-IDF + LR (b) BiLSTM + GloVe (c) DistilBERT

Figure 2: Confusion matrices for all models.



(a) BiLSTM training curves (b) DistilBERT training curves

Figure 3: Training dynamics for neural models.

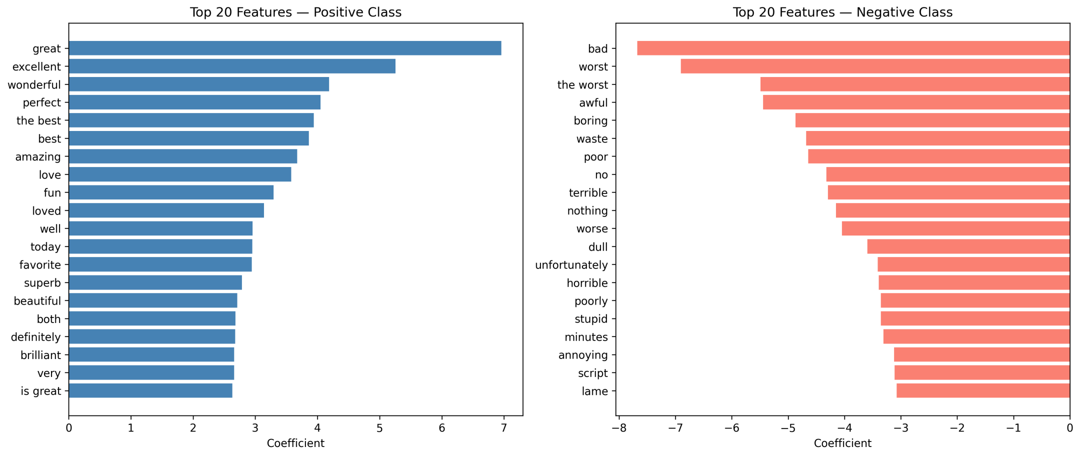


Figure 4: Top TF-IDF features for positive and negative classes.

1.6 Error Analysis

We analyze misclassifications with a focus on common error patterns. The counts are: all three models wrong (774), TF-IDF only wrong (863), BiLSTM only wrong (875), DistilBERT only wrong (512). Five representative errors from the DistilBERT test set include:

- **Negation:** reviews with localized negation often flip sentiment cues (e.g., "not like this movie").
- **Mixed sentiment:** positive and negative clauses co-occur, making the overall label ambiguous.
- **Domain terms:** film-specific words (e.g., "plot", "acting") can appear in both positive and negative contexts.
- **Sarcasm:** positive surface words mask negative intent.
- **Subtle tone:** mild criticism or praise without explicit sentiment markers.

1.7 Discussion

Sparse TF-IDF features are strong for sentiment cues and offer clear interpretability, but they struggle with long-range dependencies and nuanced phrasing. The BiLSTM benefits from sequence modeling but underperforms TF-IDF in this setup, likely due to limited capacity and the difficulty of long reviews. DistilBERT provides the best performance by leveraging contextual embeddings that capture semantics, negation, and long-distance dependencies more effectively, at the cost of higher training time and lower transparency.

2 Question 2: Named Entity Recognition

2.1 Problem Formulation

Named Entity Recognition (NER) is a sequence labeling task. Given a token sequence $x = (x_1, \ldots, x_n)$, the model predicts a BIO tag sequence $y = (y_1, \ldots, y_n)$, where each $y_i \in \{\mathtt{O}, \mathtt{B}\text{-}t, \mathtt{I}\text{-}t\}$ and $t \in \{\mathtt{PER}, \mathtt{ORG}, \mathtt{LOC}, \mathtt{MISC}\}$. We evaluate entity-level precision, recall, and F1 using the `seqeval` metric.

2.2 Dataset Description

We use the CoNLL-2003 dataset with official train/validation/test splits. The label inventory contains 9 BIO tags: `O`, `B/I-PER`, `B/I-ORG`, `B/I-LOC`, `B/I-MISC`.

Table 3: CoNLL-2003 dataset statistics.

Split	**Sentences**	**Tokens**	**Avg Len**	**Entity %**
Train	14,041	203,621	14.50	16.72
Validation	3,250	51,362	15.80	16.75
Test	3,453	46,435	13.45	17.47

Table 4: Entity token counts per split.

Split	**PER**	**ORG**	**LOC**	**MISC**
Train	11,128	10,025	8,297	4,593
Validation	3,149	2,092	2,094	1,268
Test	2,773	2,496	1,925	918

2.3 Methodology

BiLSTM-CRF. We build a word-level vocabulary (20k) and a character vocabulary, initialize word embeddings with GloVe 100d, and use a character BiLSTM to produce subword-aware representations. The final architecture is word+char embeddings $\rightarrow$ BiLSTM $\rightarrow$ linear layer $\rightarrow$ CRF. The CRF models label dependencies and enforces BIO-consistent transitions.

BERT token classification. We use `bert-base-cased` with a token classification head. Because BERT uses WordPiece tokenization, we align labels by assigning the original label to the first subword and `-100` to subsequent subwords and special tokens (ignored in the loss). We train for 3 epochs with batch size 16 and learning rate 5×10^{-5}.

2.4 Experimental Setup

We fix the random seed to 42 and evaluate once on the test split. Early stopping is applied to BiLSTM-CRF based on validation F1 (patience 5). The BERT training time is 138.23 seconds; BiLSTM-CRF timing was logged as 0.0 seconds in the output (likely a timing artifact).

2.5 Results

Table 5: Per-entity and overall results on the test set.

Entity	Model	Precision	Recall	F1
PER	BiLSTM-CRF	0.9363	0.9091	0.9225
PER	BERT	0.9645	0.9591	0.9618
ORG	BiLSTM-CRF	0.8712	0.8104	0.8397
ORG	BERT	0.8733	0.9043	0.8885
LOC	BiLSTM-CRF	0.8972	0.9107	0.9039
LOC	BERT	0.9324	0.9268	0.9296
MISC	BiLSTM-CRF	0.7356	0.7849	0.7595
MISC	BERT	0.7691	0.8162	0.7920
Overall	BiLSTM-CRF	0.8793	0.8651	0.8721
Overall	BERT	0.9024	0.9157	0.9090

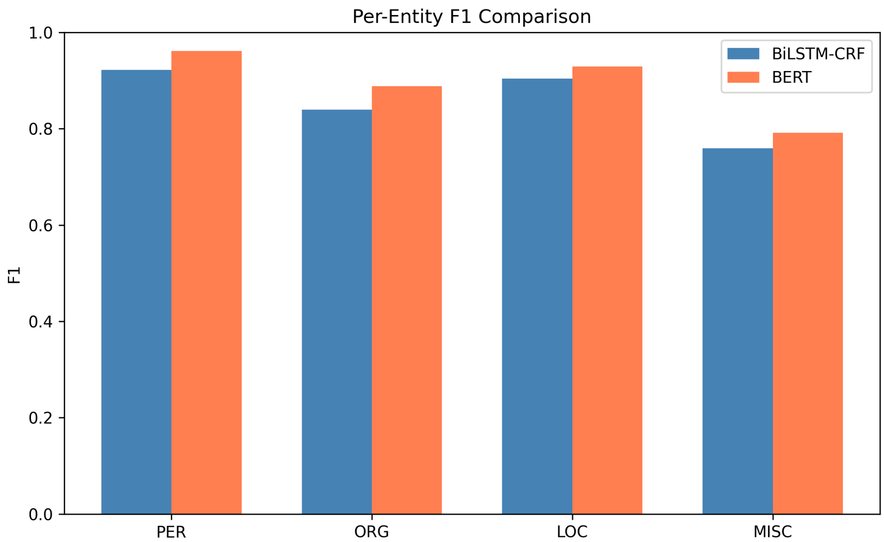


Figure 5: Per-entity F1 comparison between BiLSTM-CRF and BERT.

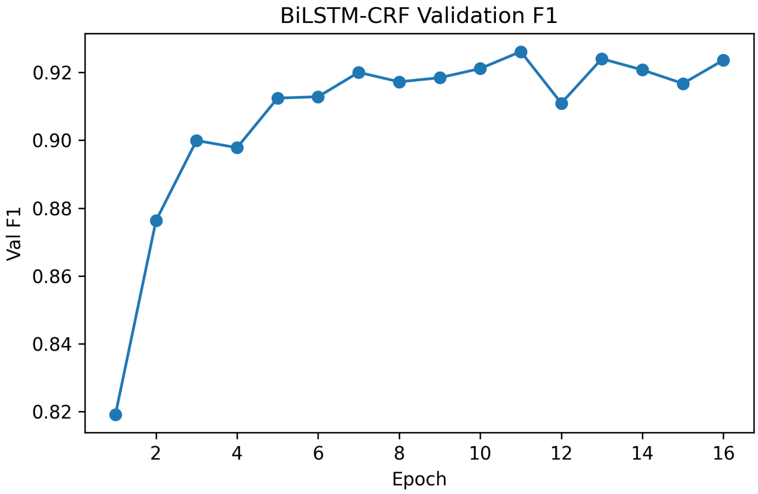
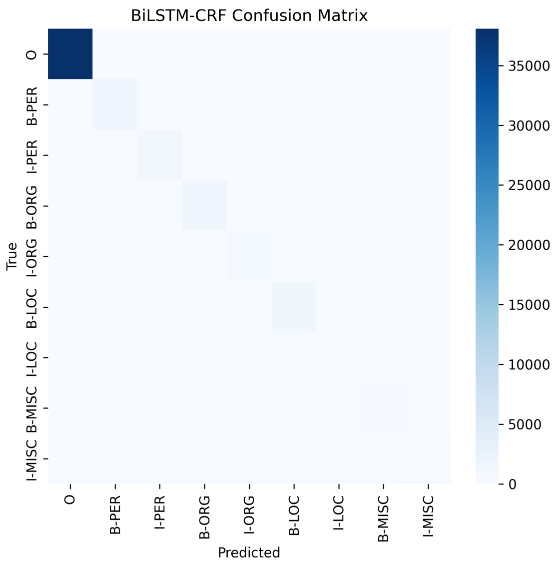
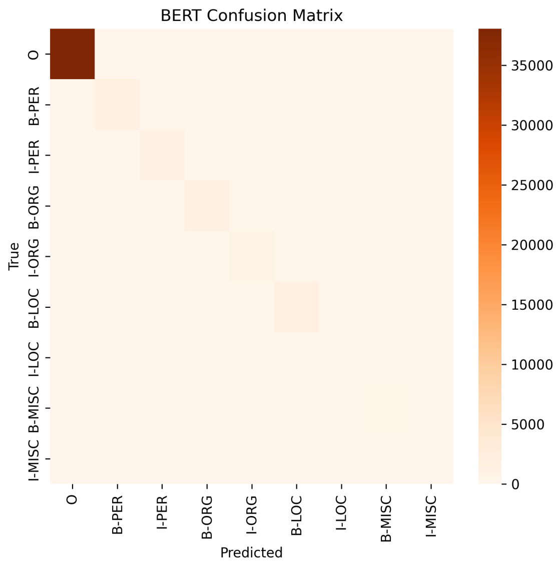


Figure 6: BiLSTM-CRF validation F1 across epochs.



(a) BiLSTM-CRF



(b) BERT

Figure 7: Confusion matrices over BIO labels.

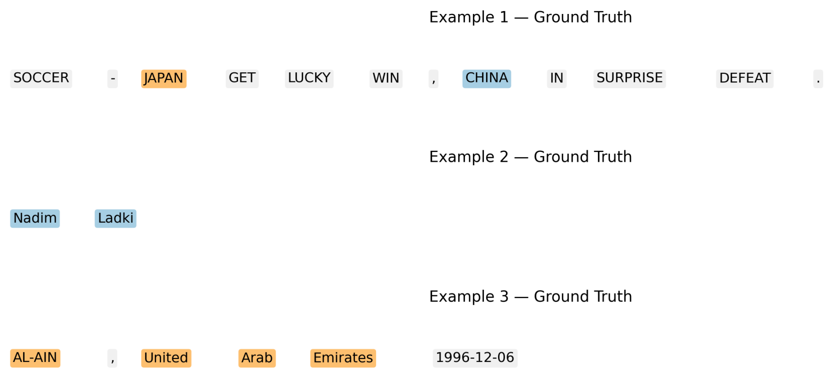


Figure 8: Example sentences with gold entity labels.

2.6 Error Analysis

We categorize errors using span-level comparisons on BERT predictions. The counts are: type confusion (317), false positives (139), boundary errors (81), and false negatives (74). Five representative errors are:

- **Type confusion (LOC → ORG/PER).** Sentence: `SOCCER - JAPAN GET LUCKY WIN , CHINA IN SURPRISE DEFEAT .`
  True: `O O B-LOC O O O O B-PER O O O O`
  Pred: `O O B-ORG O O O O B-ORG O O O O`.
  Hypothesis: country names in sports headlines are often interpreted as organizations.
- **False negative (missed PER/LOC).** Sentence: `RUGBY UNION - CUTTITTA BACK FOR ITALY AFTER A YEAR .`
  True: `B-ORG I-ORG O B-PER O O B-LOC O O O O`
  Pred: `B-ORG I-ORG O B-ORG O O O O O O O`.
  Hypothesis: person names in uppercase are confused with organizations, and nearby locations are dropped.
- **Boundary error (missing initial token).** Sentence: `Cuttitta announced his retirement after the 1995 World Cup , ...`
  True: `B-PER ... B-MISC I-MISC I-MISC ...`
  Pred: `B-PER ... O B-MISC I-MISC ...`.
  Hypothesis: multi-token event names are truncated.
- **Boundary error (invalid BIO start).** Sentence: `FREESTYLE SKIING-WORLD CUP MOGUL RESULTS .`
  True: `O B-MISC I-MISC O O O`
  Pred: `O O I-MISC O O O`.
  Hypothesis: model sometimes emits `I-` without a preceding `B-` for hyphenated entities.
- **False positive (spurious MISC).** Sentence: `CRICKET - LARA SUFFERS MORE AUSTRALIAN TOUR MISERY .`
  True: `O O B-PER O O O O O O`
  Pred: `O O B-MISC O O B-MISC O I-MISC O`.
  Hypothesis: sports-related phrases are over-labeled as `MISC`.

2.7 Discussion

BERT outperforms BiLSTM-CRF overall (0.909 vs 0.872 F1) and across all entity types, with the largest gains on PER and ORG. Both models struggle most with MISC, indicating weaker boundaries for heterogeneous categories. The dominant error type is label confusion, suggesting that headline-style contexts and uppercase tokens blur distinctions between LOC, ORG, and PER. Incorporating gazetteers or additional contextual features could reduce these confusions.

3 Question 3: Text Summarization

3.1 Problem Formulation

We study text summarization on CNN/DailyMail and compare extractive methods (selecting sentences) against abstractive methods (generating new text). Given an article x, the system outputs a summary $\hat{y}$ that should be fluent, faithful to the source, and cover the main points. We evaluate with ROUGE, BLEU, METEOR, and BERTScore.

3.2 Dataset Description

CNN/DailyMail (version 3.0.0) provides news articles with reference highlights. We use a subset for tractable computation.

Table 6: CNN/DailyMail dataset statistics.

Split	**Original Size**	**Subset Size**
Train	287,113	10,000
Validation	13,368	1,000
Test	11,490	1,000

Table 7: Average document statistics (computed on test subset).

Avg Article Len (words)	**Avg Summary Len (words)**	**Avg Sentences**	**Compression Ra**
676.08	55.21	32.91	12

3.3 Methodology

Extractive (TextRank, LexRank). Both algorithms build a sentence similarity graph and rank sentences by centrality. The summary is formed by selecting the top-k sentences (here $k = 3$). TextRank uses a PageRank-style score on sentence similarity, while LexRank adds an explicit cosine similarity graph with thresholding.

Abstractive (BART, T5). We use encoder-decoder transformers. BART-large-cnn is fine-tuned for summarization and generates fluent, condensed summaries. T5-small is prompted with a "summarize:" prefix and generates shorter, more compressed outputs.

3.4 Experimental Setup

All runs use a fixed seed (42). Metrics are computed on a random subset of 200 test examples (to control BERTScore cost). We report average inference time per article for each method.

3.5 Results

Table 8: Summarization metrics (evaluation size = 200).

Method	R-1	R-2	R-L	BLEU	METEOR	BERT F1	Len	ms
TextRank	0.3361	0.1329	0.2155	0.0728	0.3020	0.8623	96.00	62.51
LexRank	0.3501	0.1340	0.2195	0.0827	0.2806	0.8650	74.17	60.68
BART	0.4375	0.2166	0.3148	0.1467	0.3308	0.8826	61.28	898.39
T5	0.3795	0.1747	0.2683	0.1030	0.2531	0.8680	46.25	188.26

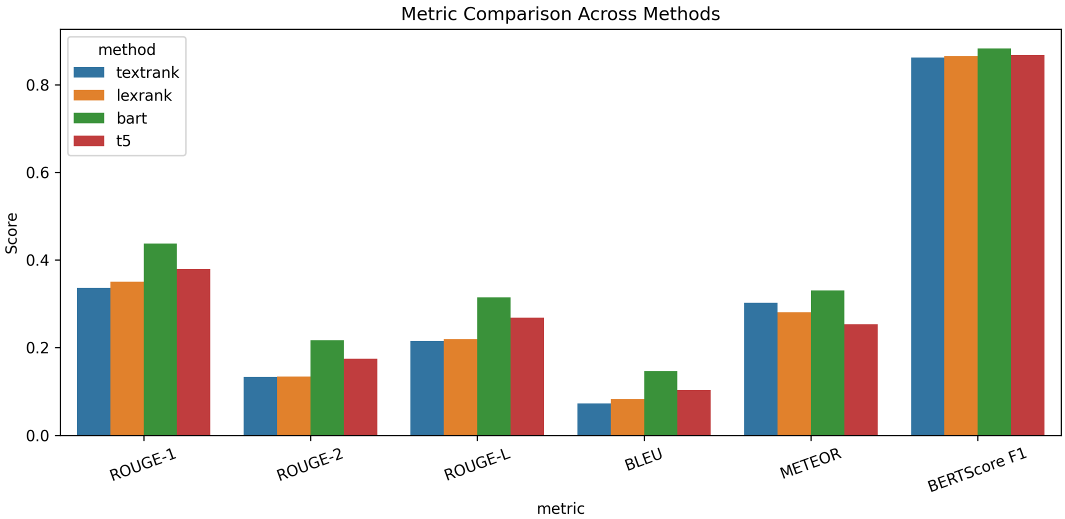


Figure 9: Metric comparison across all four methods.

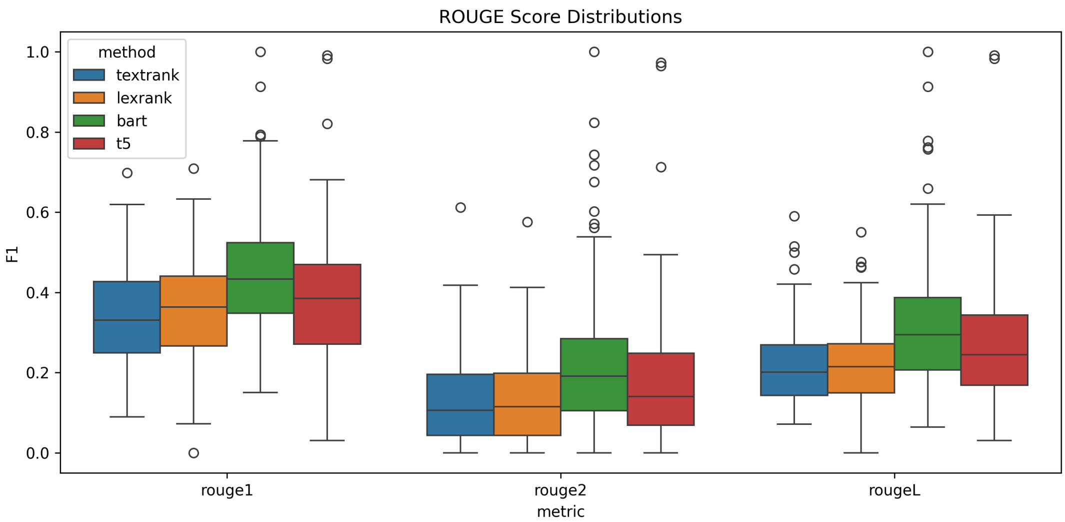


Figure 10: ROUGE score distributions for each method.

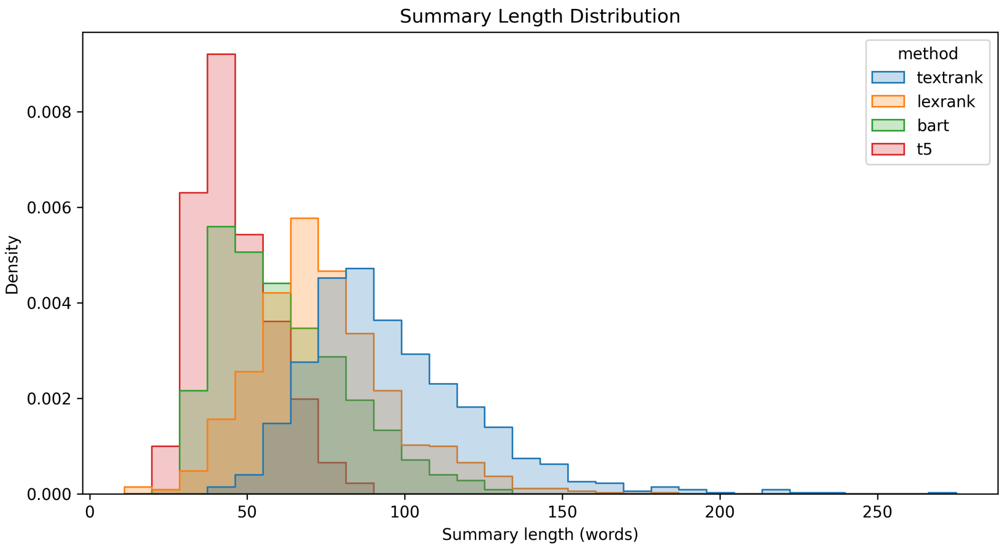


Figure 11: Summary length distributions.

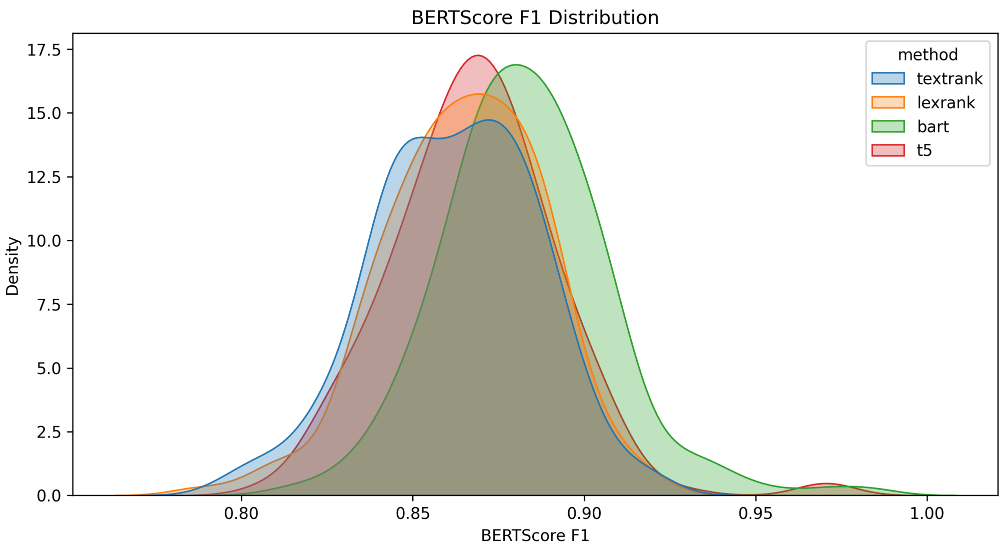


Figure 12: BERTScore F1 distributions.

3.6 Qualitative Analysis

We present three representative examples highlighting differences in fluency, faithfulness, and coverage. Text snippets are truncated for brevity.

Table 9: Example 1 (extractive wins).

Field	**Text**
Article	The legal adviser to Alison Saunders was a barrister in the same chambers as the son of Lord Janner...
Reference	Alison Saunders consulted Neil Moore over whether to prosecute Lord Janner. She said the decision was difficult and cited his health.
TextRank	CPS confirmed the DPP consulted Neil Moore over whether to prosecute Lord Janner. A spokesman said she decided alone. She said it was her job to make very difficult decisions.
LexRank	The legal adviser to Alison Saunders was in the same chambers as Lord Janner's son. A CPS spokesman said she decided alone. She admitted he should have been charged before.
BART	CPS confirmed the DPP consulted Neil Moore. Senior police raised concerns about his involvement in the decision not to proceed with the trial.
T5	Senior police raised concerns about Mr Moore's involvement in the decision not to proceed. Mrs Saunders said she made the decision on her own.

Table 10: Example 2 (abstractive wins).

Field	**Text**
Article	A British couple and their four children were believed to have fled the UK to Syria; a Turkish official claimed they were arrested in Ankara...
Reference	Asif Malik left the UK with his partner and children. A Turkish official said they were arrested in Ankara.
TextRank	Focuses on neighbors and broader context about travel to Syria, with extra background on other cases.
LexRank	States the family fled to Syria and were arrested in Turkey, but includes unrelated background.
BART	Summarizes the travel path and the claim that they were arrested in Ankara, closely matching the reference.
T5	Produces a fluent summary of the arrest and route; highly compressed and faithful.

Table 11: Example 3 (both struggle).

Field	**Text**
Article	The Obama family took an impromptu trip to Great Falls Park; reporters were sent home, sparking confusion...
Reference	The president left the White House after reporters were sent home and the family hiked at Great Falls Park for about 50 minutes.
TextRank	Extracts multiple descriptive sentences that are faithful but verbose and uneven.
LexRank	Similar to TextRank, it is faithful but misses the key timeline in the reference.
BART	Adds an unrelated detail about golfing, hurting factual precision.
T5	Focuses on the press pool confusion and the outing, but omits timing details.

3.7 Discussion: Trade-offs

Abstractive models (especially BART) achieve the strongest ROUGE and BERTScore, and their summaries are more concise and readable. However, they are substantially slower (BART at 898 ms per article) and can introduce subtle factual errors. Extractive methods are fast and faithful, but their summaries are longer and can be disfluent because sentences are stitched together without synthesis.

Metric limitations. ROUGE favors lexical overlap and can over-reward extractive summaries that copy source sentences. Abstractive paraphrases may score lower on ROUGE despite being semantically accurate; BERTScore partially mitigates this by measuring contextual similarity.

4 Question 4: Machine Translation

5 Question 5: Language Modeling

A Experimental Configuration Summary

Table 12: Global experimental settings applied to all questions.

Setting	**Value**
Random seed	42
Python version	3.10
PyTorch version	2.x
GPU	Google Colab T4
Test set usage	Once (final evaluation only)